Practice workbook · Finding an unknown constant from a condition on the roots

Using a given fact about the roots (a relation between them, or the value of a symmetric expression or difference) to form and solve an equation for an unknown coefficient.

10 question parts: 1 worked-example retry and 9 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: FPM-QUA-09

1. Question 5 · part c · 3 marks

2022 JUN · Paper 02

Worked-example retry · Attempt part c only.

5 The roots of the quadratic equation $2x^2 + (6 + 2p)x + 2p = 0$ are α and β

(a) Write down an expression in terms of p for

(i) $\alpha + \beta$ (ii) $\alpha\beta$

(2)

(b) Show that $(\alpha - \beta)^2 = 9 + 2p + p^2$

(4)

Given that $(\alpha - \beta) = 3$

(c) find the possible values of p

(3)

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1. Question 5 · part c · 3 marks

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Worked-example retry · Attempt part c only.

Question 5 continued

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1. Question 5 · part c · 3 marks

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Worked-example retry · Attempt part c only.

Question 5 continued

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1. Question 5 · part c · 3 marks

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Worked-example retry · Attempt part c only.

Question 5 continued

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(Total for Question 5 is 9 marks)

1. Question 5 · part c · 3 marks · Working and answer

Working and answer

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2. Question 11 · part a.ii · 5 marks

2019 JUN · Paper 02 (regional R)

Further practice · Attempt part a.ii only.

11 The quadratic equation $x^2 - px + q = 0$ where $p > 0$, has roots α and β.

Given that $2\alpha\beta = 3$ and that $4(\alpha^2 + \beta^2) = k^2 - 6k - 3$ where $k > 3$

(a) (i) write down the value of q,

(ii) find an expression, in terms of k, for p.

(5)

Given also that $7\alpha\beta = 3(\alpha + \beta)$

(b) find the value of k.

(2)

(c) Hence form an equation, with integer coefficients, which has roots

$$\frac{\alpha}{\alpha + \beta} \text{ and } \frac{\beta}{\alpha + \beta}$$

(5)

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2. Question 11 · part a.ii · 5 marks · Working and answer

Working and answer

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3. Question 11 · part b · 2 marks

2019 JUN · Paper 02 (regional R)

Further practice · Attempt part b only.

11 The quadratic equation $x^2 - px + q = 0$ where $p > 0$, has roots α and β.

Given that $2\alpha\beta = 3$ and that $4(\alpha^2 + \beta^2) = k^2 - 6k - 3$ where $k > 3$

(a) (i) write down the value of q,

(ii) find an expression, in terms of k, for p.

(5)

Given also that $7\alpha\beta = 3(\alpha + \beta)$

(b) find the value of k.

(2)

(c) Hence form an equation, with integer coefficients, which has roots

$$\frac{\alpha}{\alpha + \beta} \text{ and } \frac{\beta}{\alpha + \beta}$$

(5)

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3. Question 11 · part b · 2 marks · Working and answer

Working and answer

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4. Question 8 · part b · 5 marks

2023 JAN · Paper 02 (regional R)

Further practice · Attempt part b only.

8 The quadratic equation $3x^2 - kx - 1 = 0$, where k is a positive integer, has roots α and β

(a) Show that $\alpha^2 + \beta^2 = \dfrac{k^2 + 6}{9}$

(3)

Given that $\alpha^4 + \beta^4 = \dfrac{466}{81}$

(b) find the value of k

(5)

(c) Hence form an equation, with integer coefficients, which has roots

$$\frac{\alpha^3 + \beta}{\beta} \text{ and } \frac{\beta^3 + \alpha}{\alpha}$$

(6)

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4. Question 8 · part b · 5 marks · Working and answer

Working and answer

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5. Question 10 · part a · 8 marks

2024 JUN · Paper 01 (regional R)

Further practice · Attempt part a only.

10 The quadratic equation $2x^2 + kx + 4 = 0$ has roots α and β such that

$$k < 0 \text{ and } \alpha > \beta$$

Given that $\alpha^2 - \beta^2 = \frac{7\sqrt{17}}{4}$

(a) show that $k = -7$

(8)

(b) Hence form a quadratic equation that has roots

$$(\alpha - \beta) \text{ and } (\alpha + \beta)$$

(4)

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5. Question 10 · part a · 8 marks · Working and answer

Working and answer

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6. Question 7 · part a.i · 0 marks

2022 JAN · Paper 02

Further practice · Attempt part a.i only.

7 $f(x) = 2x^2 + px + q$ where p and q are integers
$g(x) = 14x^2 + 37x + 14$

The equation $f(x) = 0$ has roots α and β

The equation $g(x) = 0$ has roots $\frac{\alpha}{\beta}$ and $\frac{\beta}{\alpha}$

Given that $p + q = -4$ where $p > 0$ and without solving the equation $g(x) = 0$

(a) find

(i) the value of p

(ii) the value of q

(9)

Given also that $\alpha > \beta$

(b) show that $\alpha^2 - \beta^2 = -\frac{3\sqrt{65}}{4}$

(4)

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6. Question 7 · part a.i · 0 marks · Working and answer

Working and answer

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7. Question 7 · part a.ii · 9 marks

2022 JAN · Paper 02

Further practice · Attempt part a.ii only.

7 $f(x) = 2x^2 + px + q$ where p and q are integers
$g(x) = 14x^2 + 37x + 14$

The equation $f(x) = 0$ has roots α and β

The equation $g(x) = 0$ has roots $\frac{\alpha}{\beta}$ and $\frac{\beta}{\alpha}$

Given that $p + q = -4$ where $p > 0$ and without solving the equation $g(x) = 0$

(a) find

(i) the value of p

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(9)

Given also that $\alpha > \beta$

(b) show that $\alpha^2 - \beta^2 = -\frac{3\sqrt{65}}{4}$

(4)

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7. Question 7 · part a.ii · 9 marks · Working and answer

Working and answer

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8. Question 8 · part whole · 11 marks

2023 JAN · Paper 02

Further practice · Attempt part whole only.

8 The quadratic equation

$$x^2 - 4k\sqrt{2}x + 2k^4 - 1 = 0$$

where k is a positive constant, has roots α and β

Given that $\alpha^2 + \beta^2 = 66$ and that $\alpha^3 + \beta^3 = p\sqrt{2}$ where p is an integer,

find the value of p

(11)

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8. Question 8 · part whole · 11 marks

2023 JAN · Paper 02

Further practice · Attempt part whole only.

Question 8 continued

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8. Question 8 · part whole · 11 marks

2023 JAN · Paper 02

Further practice · Attempt part whole only.

Question 8 continued

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8. Question 8 · part whole · 11 marks

2023 JAN · Paper 02

Further practice · Attempt part whole only.

Question 8 continued

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(Total for Question 8 is 11 marks)

8. Question 8 · part whole · 11 marks · Working and answer

Working and answer

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9. Question 7 · part b · 4 marks

2025 NOV · Paper 01

Further practice · Attempt part b only.

7 (a) Show that $(\alpha+\beta)(\alpha^2-\alpha\beta+\beta^2)=\alpha^3+\beta^3$

(1)

The equation $2x^2+8x-k=0$ has roots α and β and where k is a constant

Given that $\alpha^3+\beta^3=-94$

(b) show that $k=5$

(4)

Given that $\alpha>\beta$ and without solving the equation $2x^2+8x-k=0$

(c) (i) show that $\alpha-\beta=\sqrt{26}$

(3)

(ii) hence find the exact value of $\alpha^3-\beta^3$

(2)

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9. Question 7 · part b · 4 marks · Working and answer

Working and answer

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10. Question 6 · part a · 5 marks

2023 JUN · Paper 02 (regional R)

Further practice · Attempt part a only.

6 $f(x) = 2x^2 + 5x - p$

The equation $f(x) = 0$ has roots α and β

Given that $\alpha^3 + \beta^3 = -\frac{215}{8}$

(a) find the value of p

(5)

Without solving the equation $f(x) = 0$

(b) form a quadratic equation, with integer coefficients, that has roots

$$\frac{\alpha + \beta}{\alpha^2} \text{ and } \frac{\alpha + \beta}{\beta^2}$$

(5)

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10. Question 6 · part a · 5 marks · Working and answer

Working and answer

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